

App Dev Project Report

1. Student Details

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About Me: I am a student at IIT Madras BS Degree program. I am passionate about learning new technologies. I enjoy building meaningful applications and improving my skills.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a **Placement Portal Application** web application that allows **Admin (Institute), Company, and students** to interact with the system based on their roles.

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows users to efficiently manage campus recruitment activities involving companies, students, and placement drives.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in identifying errors during debugging, used to clarify programming concepts, and provide minor code suggestions when needed. AI was also used to help improve the wording and formatting of some sections of the project documentation.

The extent of AI/LLM usage is around **10–15%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging decisions, and integration were done manually.

4. Technologies and Frameworks Used

Technology Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask Session	Managing logged-in users
SQLite	Database for storing user data
Python	Backend programming language
HTML & CSS	Structure & Styling of web pages

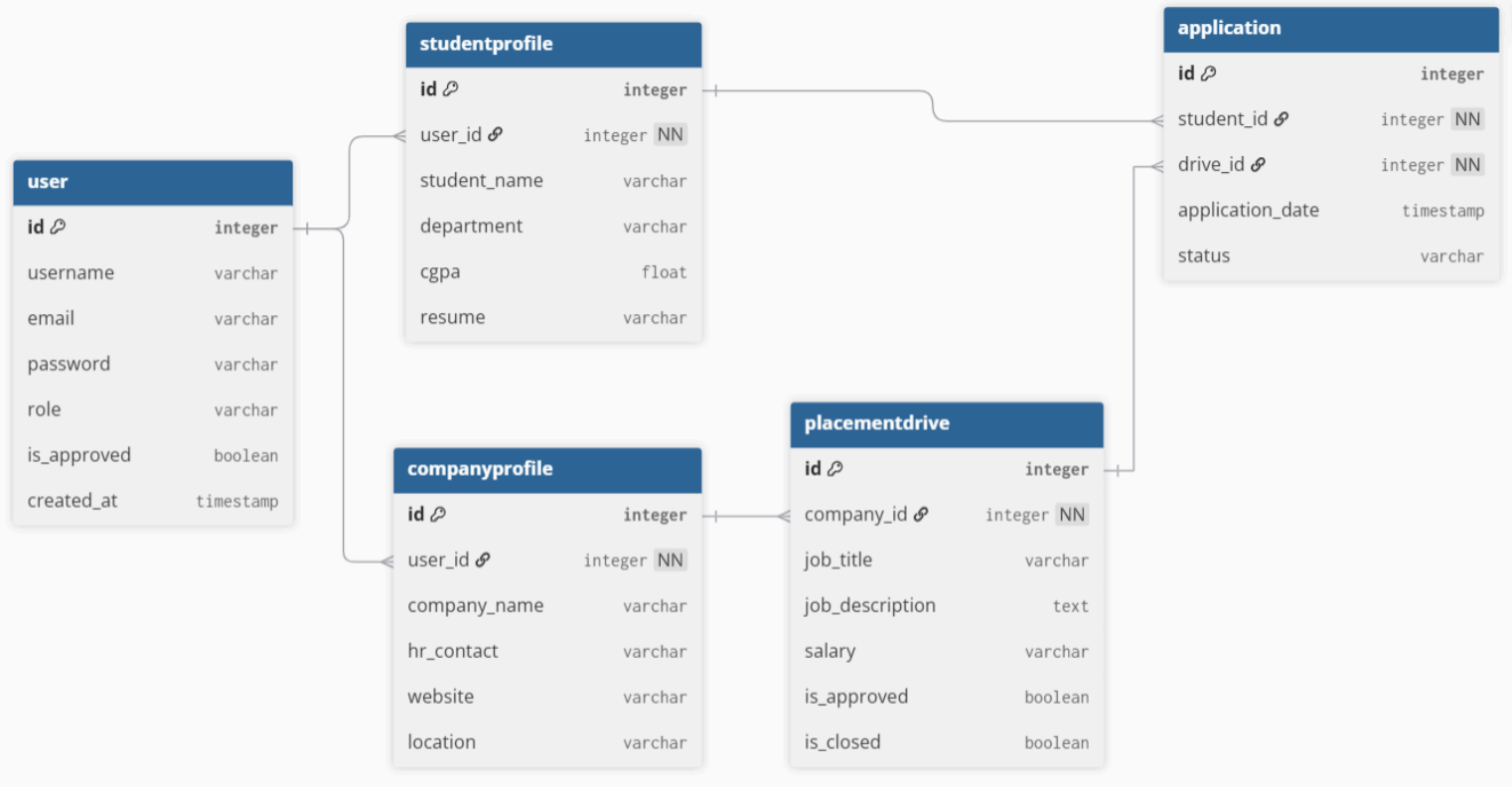
5. Database Schema / ER Diagram

Tables:

1. **User** — stores user account information (id, username, email, password, role, is_approved, created_at)
2. **Studentprofile** — stores student profile details(id, user_id, student_name, department, cgpa, resume)
3. **Companyprofile** — stores company profile details(id, user_id, company_name, hr_contact, website, location)
4. **Placementdrive** — stores job posting details(id, company_id, job_title, job_description, salary, is_approved, is_closed)
5. **Application** — stores all applications applied by students for placement drives (id, student_id, drive_id, application_date, status)

Relationships:

- One-to-One → User → Studentprofile
- One-to-One → User → Companyprofile
- One-to-Many → Companyprofile → Placementdrive
- One-to-Many → Studentprofile → Application
- One-to-Many → Placementdrive → Application



6. API Resource Endpoints

a. Common Routes

Endpoint	Method	Description
/	GET	Index / Home page
/registration	GET, POST	Register new user (student/company)
/login	GET, POST	User login (Student / Company / Admin)
/logout	GET	Logout user and clear session

b. Student Module

Endpoint	Method	Description
/studentdash	GET	View student dashboard with companies and drives
/student_profile	GET, POST	Create or update student profile
/stu_com_overview/<company_id>	GET	View company overview and available drives
/stu_com_view/<drive_id>	GET	View details of a specific job drive
/student_history	GET	View student's application history
/apply_job/<drive_id>	GET	Apply for a placement drive

c. Company Module

Endpoint	Method	Description
/companydash	GET	View company dashboard
/company_profile	GET, POST	Create or update company profile
/create_drive	GET, POST	Create new placement drive
/view_drives/<drive_id>	GET	View applications for a drive
/review-application/<app_id>	GET, POST	Review and update status of student application
/company_close/<drive_id>	GET	Close a placement drive
/company_open/<drive_id>	GET	Reopen a placement drive
/company_edit/<drive_id>	GET, POST	Edit placement drive
/company_delete/<drive_id>	GET	Delete placement drive
/whole_companystudent_profile/<student_id>/<company_id>	GET	View student profile for company
/company_stu_history/<student_id>/<company_id>	GET	View student's application history for particular company

d. Admin Module

Endpoint	Method	Description
/admindash	GET	Admin dashboard showing students, companies, drives
/admin_view_stu/<app_id>	GET	View student application details
/admin_view_com/<drive_id>	GET	View company drive details
/company_approve/<company_id>	GET	Approve company registration
/company_blacklist/<company_id>	GET	Blacklist company
/student_approve/<student_id>	GET	Approve student account
/student_blacklist/<student_id>	GET	Block student account
/drive_approve/<drive_id>	GET	Approve placement drive
/drive_disapprove/<drive_id>	GET	Disapprove placement drive
/whole_student_profile/<student_id>	GET	View complete student profile
/admin_stu_history/<student_id>	GET	View student's application history
/whole_company_profile/<company_id>	GET	View full company profile and drives

e. Navigation / View Helper Routes

Endpoint	Method	Description
/admin_indirectview_student/<app_id>	GET	Admin view student via application
/company_indirectview_student/<app_id>/<company_id>	GET	Company view student via application
/admin_mainpageview_student/<app_id>	GET	Admin view student from dashboard

7. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1CicbbmVmbixgtboi8nWyH-Cm3m-IRKmw/view?usp=sharing>
